温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Course Introduction & House Rule

Dawei Chen

Week 1-1, 02 Sep 2025

About Me

Dawei CHEN

- **Education**
  - B.S. in Economics – SEM, THU
  - Ph.D. in Information Systems – SOC, NUS
  - Postdoc in AI Governance – AISG, NUS
- **Research:** Information Privacy, Economics of GenAI
- **Interests:** Fitness, Hiking, Marathon, Scuba Diving
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- **Office Hours:** Mon & Wed 4pm – 6pm | Fri 4pm – 5pm

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Tell Me About You

- Name
- Degree program & year
- Relevant interests & experiences

(e.g., internships, projects, or extracurricular activities)

- What makes you interesting / unique?

(share something unique about yourself or your journey)

Your Knowledge on DA/BA

- Which data analysis tools are you familiar with?

(e.g., Excel, SQL/MySQL, Python, R, Tableau/PowerBI, Spark/Hadoop, etc.)

- What types of analysis have you performed?

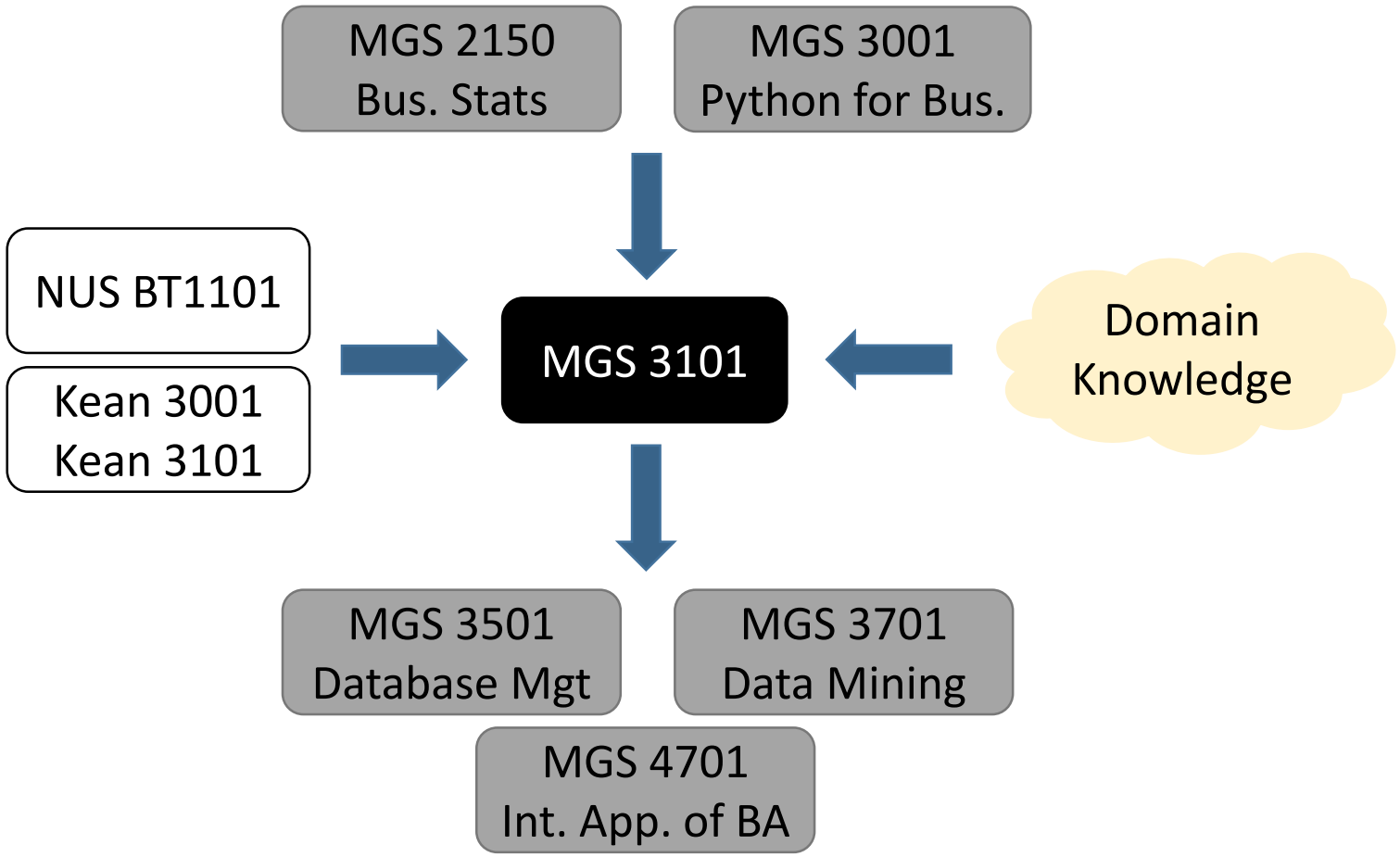
(e.g., descriptive statistics, visualization, ANOVA, regression, advanced models, etc.)

- Do you have experience with business analytics projects?

- What do you hope to learn from this course?

Heterogeneity

Course Overview (Redesign Process)



Q1: 业界一般需要做什么类型的数据分析？用什么models？这个可能取决于context和数据类型。

Ans: 我们做关于geospatial的多一些。因为我们公司跟xxx的项目多一些所以处理people movement数据多一点。 models的话clustering, regression, ML, time series都会用。

Q2: 在数据分析过程中， 一般要处理什么样的问题？代码或者数据问题。

Ans: trend, correlation, causalit, what insight you can get from data.

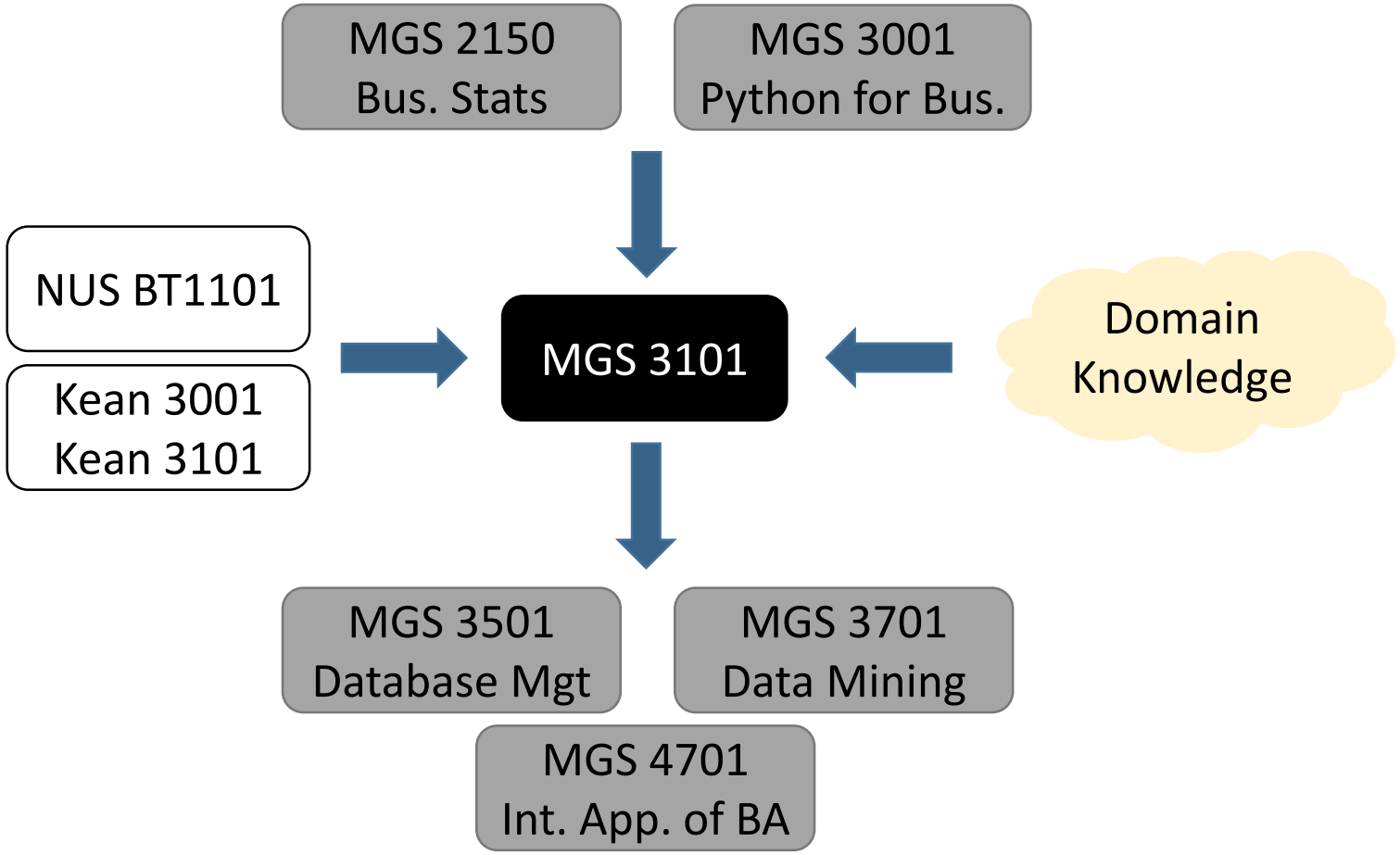
Q3: 在applied models之前， 如何进行数据整理、清洗、可视化了解数据?

Ans: 一般就是从csv或者excel或者从AWS s3 extract data到python/sql里处理。整理清洗就是调一调format比如时间的格式，geometry的格式，str，float。还会用merge, conca, filter比较多。可视化python package matplotlib, dashboard: power bi, looker studio。

Q4: 一门课data analytics with Python，一半时间讲理论， 一半时间现场演示与实际操作with jupyter notebook，一般建议讲什么topics比较合适?

Ans: introduction to jupyter notebook, numpy, pandas, matplot, data upload, cleaning, basic data statistics和visulation to understand data, applied specific models. 然后老师再弄一个group project哈哈哈哈

Course Overview (Redesign Process)



- High level
- Overview of BA

- Entry level DA
- Hand-on DA
- Internship
- Senior Thesis

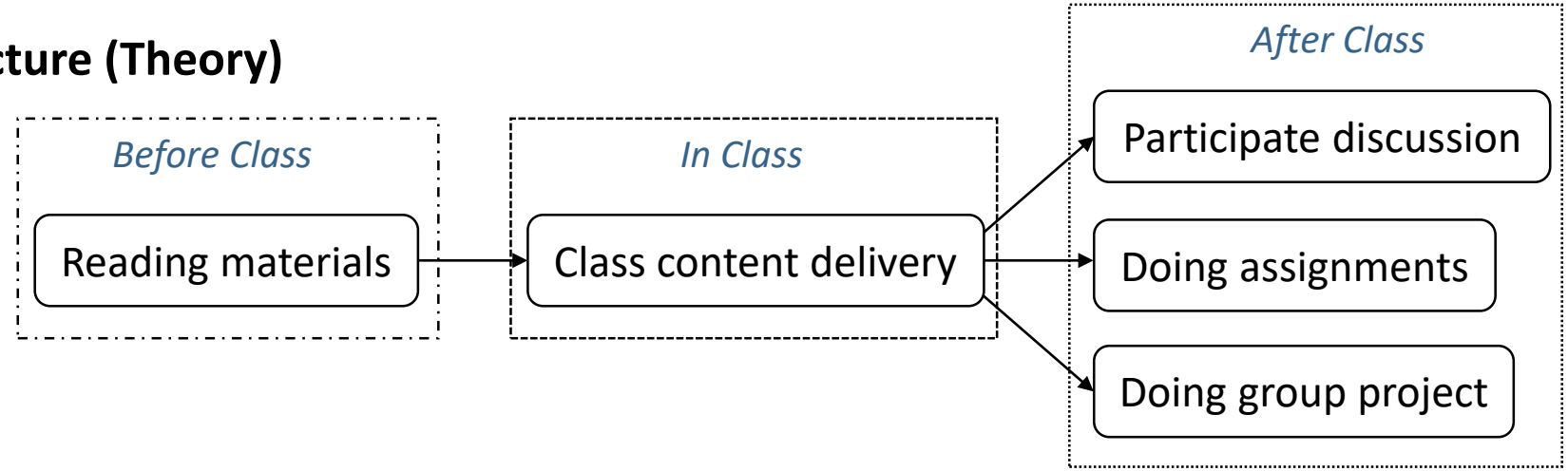
Learning Outcomes

Upon successful completion of this course, the student will be able to:

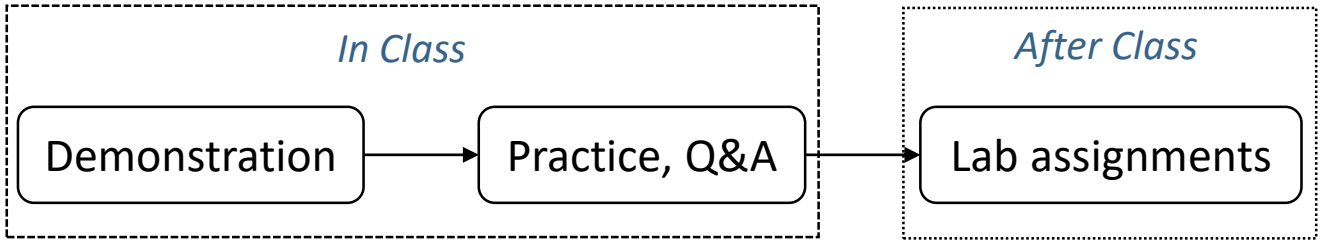
- Demonstrate the motivation, drivers, and importance of business analytics
- Describe the different types of **analytics** and illustrate the objectives of each one of them
- Discuss the importance of **data visualization** and apply different types of visualization techniques using **Python**
- Identify some of the emerging technologies that may impact analytics, business intelligence, and data science, such as Internet of Things (IOT), cloud computing, **artificial intelligence**
- Outline the ethical and legal issues related to deployment of analytics
- Identify main characteristics, **skills**, and requirements of analytics-related **career** opportunities
- *Able to **independently** complete **beginner-level** business analytics projects (minimum)*
- ***Communicate** the results of a business analytics project to a non-technical audience of decision maker and stakeholders*

Course Format

I. Lecture (Theory)



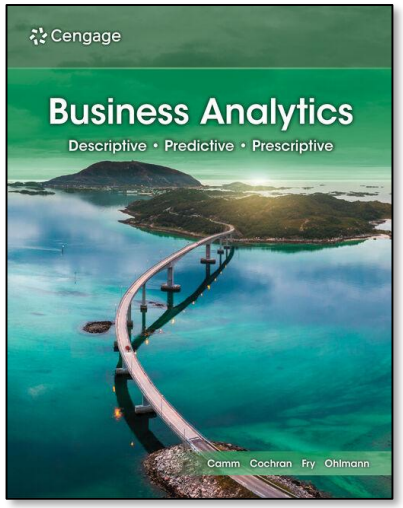
II. Lab (Practice) (>50%)



Please bring your laptop!

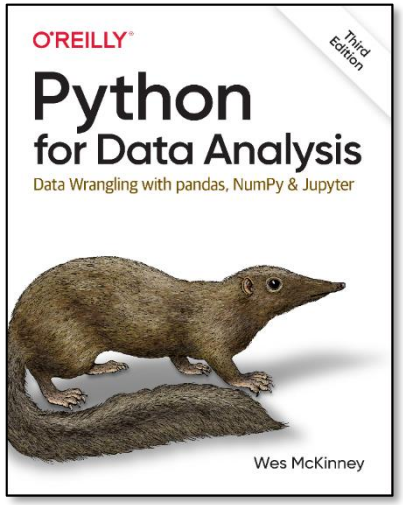
Course Materials

Theory



Business Analytics, 5th Edition, by Camm et al., Cengage 2023.

Practice



Python for Data Analysis, 3rd Edition, by Wes Mckinney, O'Reilly 2022.

**Additional reading materials may be introduced throughout the semester.*

Course Schedule (v1.0)

Date		Topics	Assignments
W1	09/02	Course Introduction & House Rule	
	09/04	Introduction to Business Analytics (Camm, Ch1)	
W2	09/09	(lab) Jupyter Notebook, Data Loading and Storage, NumPy and Pandas	
	09/11	Descriptive Statistics (Camm, Ch2)	
W3	09/16	(lab) Descriptive Analysis with Python	
	09/18	Data Wrangling: Data Management and Cleaning Strategies (Camm, Ch4)	
W4	09/23	(lab) Data Wrangling, Cleaning and Preparation (Wes, Ch7 and Ch8)	
	09/25	Data Visualization (Camm, Ch2)	
W5	09/30	(lab) Plotting and Visualization (Wes, Ch9)	Assignment 1: 09/26
	10/02	(no class)	
W6	10/07	(no class)	
	10/09	(lab)	Assignment 2: 10/10
W7	10/14	Review, Q&A for Midterm Exam	
	10/16	Midterm Exam (computer-based)	
W8	10/21	Statistical Inference (Camm, Ch7)	
	10/23	(lab)	

**Tue & Thu 5:30pm-6:45pm. Class in white is in CBPM A-202; Class in grey is in SLAC 301.*

Course Schedule (v1.0)

Date		Topics	Assignments
W9	10/28	Simple Linear Regression (Camm, Ch8)	
	10/30	(lab)	Assignment 3: 10/31
W10	11/04	Multiple Linear Regression	
	11/06	(lab)	
W11	11/11	Logistic Regression	
	11/13	(lab)	Assignment 4: 11/14
W12	11/18	Time Series Analysis (Camm, Ch9)	
	11/20	(lab)	
W13	11/25	Mediation and Moderation Analysis	
	11/27	(lab)	
W14	12/02	Panel Data Analysis	
	12/04	(lab)	
W15	12/09	Guest Speaker (online)	
	12/11	Group Project Presentation	
W16	12/16	Review, Q&A for Final Exam	
	12/18	Final Exam (written)	

**Tue & Thu 5:30pm-6:45pm. Class in white is in CBPM A202; Class in grey is in SLAC 301.*

Course Assessment

Grade Breakdown	
Participation	10
Assignments	20
Midterm Exam(computer-based)	25
Final Exam (written)	25
Group Project	20
Total	100

Responsible AI Adoption

Negative impact of over-reliance on AI

- Efficiency (short term) vs. critical thinking damage (long term) (Kosmyna et al. 2025)
- Loss of fundamental understanding, skill erosion, team knowledge gap
- Good taste is more important than ever (Nitin Nohria 2025)

Suggestion

- Avoid over-reliance on AI
- AI as a complement, not a replacement
- Master prompt engineering
- Verify AI output
- Ensure proper citation and attribution

Q&A

CBPM B214

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Mon & Wed 4pm – 6pm | Fri 4pm – 5pm

WeChat group

Join Private Group 面对面建群

0902